

Shared exclusions PID audit

Draft scope, exact checks and application limits

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This report reviews categorical shared-exclusions PID and its extensions through three evidence channels: defining sources, formal and numerical correspondence, and statistical and application assumptions. The current bounded checks agree. Two source/navigation errors and several diagnostic weaknesses were found and preserved. Full triple-check completion remains open. No scientific-priority, generic calibration or deployment-benefit claim is made.

Object and assumptions

The categorical object is the finite-law shared-exclusions measure of [Makkeh, Gutknecht and Wibral, version 5](#). For source indices $1, \dots, n$, an antichain α is a nonempty family of nonempty source groups, with no group containing another. At a supported observation $z = (s_1, \dots, s_n, t)$, define

$$A_\alpha(z) = \bigcup_{a \in \alpha} \bigcap_{i \in a} \{S_i = s_i\}.$$

Thus a group uses AND; the family uses OR. Write $u = P(A_\alpha)$, $j = P(A_\alpha, T = t)$ and $q = P(T = t)$. Support gives $0 < j \leq \min(u, q)$. The cumulative components, in nats, are

$$c^+ = -\log u, \quad c^- = \log(q/j), \quad c^{\text{net}} = \log \frac{j}{uq}.$$

For $\beta \preceq \alpha$ when each group in α contains a group in β , each component's pointwise atom vector $\pi(z)$ solves $c_\alpha(z) = \sum_{\beta \preceq \alpha} \pi_\beta(z)$. Finite Möbius inversion solves this linear system. For $p(z) = P(Z = z)$, define averaged net atoms $\Pi_\alpha = \sum_{z: p(z) > 0} p(z) \pi_\alpha^{\text{net}}(z)$. Informative and misinformative component nonnegativity does not imply net nonnegativity. Negative net atoms must remain visible.

Categorical MGW, continuous Ehrlich, Williams--Beer $I_{\min}$ and KSG are different objects. Quantization, projection and added noise change variables or laws. A support declaration is an assumption, not a property inferred from unique observed rows. The continuous full PID3 path remains research-only.

The finite reference calculations use explicitly specified laws and exact rational arithmetic or exact rational log-products. Floating replicas are labeled separately. Where rows are called IID, complete source-and-target rows are independent and identically distributed; coordinates within a row need not be independent. Files are assumed stable during bounded reads. Hash equality is not authenticity, atomicity or independent custody. Python mode repeats share code. Model critiques are advisory, not independent human review. Decimal illustrations are not directed logarithm enclosures.

Audit coverage

The current integration catalog has 75 methods and 48 references. All novelty fields are none. Its generated rendering is coherent; this does not verify all methods or all references. The retained

primary checkout is older and dirty. It must not replace the integration or be reset without preservation.

The foundational result map has nine families. The following are current boundaries, not new adjudications of every proof.

Result family	Retained result	Important open edge
Categorical foundations	Event semantics and exact negative witnesses	Complete paper-to-code correspondence
Finite plug-in convergence	Fixed finite support and IID or stationary ergodic limits	Floating/runtime refinement
Support-change continuity	Analytic finite-alphabet modulus and scoped witnesses	Complete formal logarithmic/lattice path
Dependency-color concentration	Bounds under the declared dependency graph	Automatic valid coloring and application calibration
Exact PID2 assurance	Supplied-count two-source identities; conditional certifier	General compiled-code refinement and integration
SxPID3 factorization	Informative components factor through the complete joint source marginal; bounded audit	Full three-source certificate
Binary64 and quantizers	Measure-specific arithmetic and diagnostics	Full numerical and fitting/evaluation guarantees
KSG harmonic arithmetic	Scoped exact integer/harmonic core	Publication integration; old C12 route permanently failed
Manifold small-ball bridge	Conditional common-radius population lemma	Adaptive estimator, expectation and mixed-support transfer

Seven companions cover prefix mean, bias, gradient, support-change cusp, target-copy synergy, fixed-world comparison and first-hit cancellation. Local acceptance records are separate from full replay and publication. Later log-score, conditional Bayes, fault and availability results do not prove that a learned predictor, sensor or neural intervention improves.

Fresh formal-source joins match 11 aggregate source pins, 26 target-copy and fixed-world modules, and 83 selected source/receipt bindings. Nine auxiliary SMT snapshots match their checker pins. These mathematical packets were not rerun with a Lean kernel or SMT solver. Not all 246 aggregate theorem bodies or complete historical streams have received a fresh line-by-line review. The finite fault argument below was independently reconstructed from its complete definitions and proof.

The current full SxPID3 certificate is still proposed: Programs A--E are zero of five closed. Nineteen Fin3 propositions have been stated; no candidate proof exists. Revision-six setup failed: a negative proof-definition fixture reached Lean's built-in proposition linter before the intended declaration-kind guard. That failed campaign is preserved. A separately frozen revision-seven uses a same-type abbreviation as the negative fixture. Both actual Python modes completed setup, fresh kernel replay of the True-only smoke and six intended causal refusals. Each campaign made 23 recorded calls, including parser controls; the summary alone lists 22. No linter option, compiler flag, kernel, target, judge or estimator changed. This operational prerequisite supplies no Fin3 proof, source/cache equivalence, statistical calibration or application evidence.

Exact finite numerical checks

Generated carriers have 4, 18 and 166 nodes for two, three and four sources. The checks cover every entry of both matrix inverse products, reverse-DNF order, and all 2, 6 and 24 source permutations. The generated three-source carrier matches the literal Rust registry. These are executed finite comparisons, not native kernel proofs or arbitrary-input Rust refinement.

Binary empirical laws	Tables	Supported anchors	Maximum total count
Two sources and target	494	1,320	4

Binary empirical laws	Tables	Supported anchors	Maximum total count
Three sources and target	968	2,448	3

The corpus checks 148,032 pointwise and 58,200 averaged component expressions using exact rational log-products. Component nonnegativity holds in this corpus; 150 two-source and 336 three-source averaged net coordinates are negative. Four-source probability values and arbitrary alphabets were not exhausted.

An independent scaled-integer calculation matches all 561 committed sum-oracle expectations, including 71 exact cancellations and 11 rounded overflows. Exact Fraction arithmetic checks the direct/range identity on 6,920 small- n KSG harmonic rows and 354 declared endpoint zeros. The stress corpus has 1,278 rows, including 114 checked endpoint zeros. Decimal accuracy of the remaining 1,164 nonzero stress rows was not checked. These comparisons do not execute the Rust arithmetic helpers.

Rust implementation and computational cost

Existing categorical Rust APIs implement two to four sources; fitted quantization targets the fitted categorical law. The new prefix and availability diagnostics are Python references, not new Rust learners or estimator refinements. A case at horizon three enumerates 4,096 complete words; four-source carrier generation searches all 32,767 nonempty families of nonempty source subsets. These are finite workload counts, not measured performance. This audit reports no fresh timing, online suitability or complete allocation guarantee. Continuous brute-force neighbor paths remain quadratic in row count.

A nonempty expression can be exactly zero

In binary cell order (S_1, S_2, T) , use counts $(0, 0, 1, 1, 1, 4, 1, 0)$, total eight. For each supported anchor, compute its target count, singleton count and union count, form the two net cumulatives, subtract the union cumulative from the first singleton cumulative, and average with count/eight weights. For counts N, U, J, Q (total, event, event-target intersection and target), the log argument is $NJ/(UQ)$. At anchor 010, the singleton has $U = 2, J = 1, Q = 3$ and gives $8/(2 \cdot 3) = 4/3$; the union has $U = 3, J = 2, Q = 3$ and gives $16/9$. The full supported-row calculation is:

Anchor	Count	First singleton ratio	Union ratio
010	1	4/3	16/9
011	1	4/5	8/15
100	1	8/9	8/9
101	4	16/15	16/15
110	1	8/9	1

The 100 and 101 rows cancel within their singleton-minus-union differences. The other rows give five nonzero log terms. Clearing the denominator yields

$$8V = \log(4/5) + \log(8/9) + \log(4/3) - \log(8/15) - \log(16/9).$$

The numerator product is $128/135$; the denominator product is also $128/135$. Hence $8V = \log 1 = 0$, so $V = 0$. A nonempty canonical term list cannot prove a nonzero value. This reconstructs a retained completeness counterexample; it is not a new priority claim or a freshly certified interval.

Python logarithm replicas also produce tiny nonzero values at exact zero and tiny negative values for exact nonnegative components. Their largest recorded zero residuals are about 2.22×10^{-16} and 1.11×10^{-16} nats. These are replica observations, not measured Rust defects. A final exact sum

cannot undo rounding in upstream logarithms. Do not clamp atoms or change a kernel from these observations. Exact products or certified intervals are needed when an exact sign matters.

Prefix means and score derivatives

For the one-source block, let $Z_0 = (S_0, T_0)$ be the anchor and $Z_1, \dots, Z_h$ the later rows. Define $D_k = \mathbf{1}\{S_k \neq S_0\}$ and $R_j = \sum_{k=1}^j \mathbf{1}\{T_k = T_0\}$, the number of retained target matches. The native block is

$$B_h = \sum_{j=1}^h \left[\frac{\prod_{k=1}^j D_k}{j} - \mathbf{1}\{T_j = T_0\} \frac{\prod_{1 \leq k \leq j: T_k = T_0} D_k}{R_j} \right].$$

The second term is defined as zero when $R_j = 0$; an empty product is one. It contributes only on a current target-match arrival. Thus $B_1 = \mathbf{1}\{S_1 \neq S_0, T_1 \neq T_0\}$. Raw time j and retained rank R_j are different. Direct enumeration and a separate event-mass polynomial agree for three two-source laws, horizons zero through three, and all four atoms: 14,040 complete words, including zero-mass words. Only one law tests a nonzero tangent.

For three IID complete rows of independent fair bits S and T , at horizon two, each anchor gives the same calculation. The positive mean is $1/2 + (1/2)^2/2 = 5/8$. Negative contributions are first arrival $1/4$, a target miss followed by a target match and source miss $1/8$, and two retained misses $1/16$ divided by rank two. Thus

$$EB_2 = \frac{5}{8} - \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{32} \right) = \frac{7}{32}.$$

Replacing $R_2 = 1$ by raw rank two halves the $1/8$ target-miss/arrival subtraction, giving $7/32 + 1/16 = 9/32$. Recounting an earlier exclusion when the second row is not an arrival adds a $1/8$ subtraction, giving $7/32 - 1/8 = 3/32$. Both incorrect rules are retained as causal controls.

For a differentiable normalized positive finite law p_θ and a parameter-independent block B on $h + 1$ IID rows, write $s_\theta(z) = p'_\theta(z)/p_\theta(z)$. Differentiating the finite sum, then each product, gives

$$\frac{d}{d\theta} E_\theta B = \sum_w B(w) \sum_{r=0}^h p'_\theta(w_r) \prod_{k \neq r} p_\theta(w_k) = E_\theta \left[B \sum_{r=0}^h s_\theta(Z_r) \right].$$

The score sum includes the random anchor $r = 0$. At one source, horizon one, use diagonal masses $\theta/2$ and off-diagonal masses $(1 - \theta)/2$. The block is one exactly when both bits differ between the two rows. Its mean is $(\theta^2 + (1 - \theta)^2)/2$, whose derivative is $2\theta - 1$. At $\theta = 3/4$ the full derivative is $1/2$; omitting the anchor score gives $1/4$ by exchangeability of the two IID rows.

Equal marginals do not justify an IID second moment

At $\theta = 3/4$, the scores in binary order are $(4/3, -4, -4, 4/3)$ and $Es^2 = 16/3$. Instead of IID rows, take $Z_1 = (1 - S_0, 1 - T_0)$. Complement preserves the displayed law, so both marginals remain correct. Both bits differ, hence $B_1 = 1$; the two scores are equal. Consequently

$$E \left[(B_1(s(Z_0) + s(Z_1)))^2 \right] = 4Es^2 = 64/3.$$

For IID rows, normalization gives $Es = \sum_z p'_\theta(z) = 0$. Since $|B_1| \leq 1$, independence gives $E[B_1^2(s_0 + s_1)^2] \leq E[(s_0 + s_1)^2] = 2Es^2$. This IID unit-range upper bound is $32/3$ and is exceeded. It is not the actual IID block-score moment: that value is $2(3/8)^2(8/3)^2 + 2(1/8)^2(8)^2 = 4$. The

weaker envelope 32 is not exceeded. Identical rows give block zero and fail as an attack; that failed construction remains archived. The successor diagnostic constructs the actual complement pairs, checks both marginals and evaluates the native block. This is a finite counterexample to dropping IID, not a confidence-coverage or optimization theorem.

Complete-key return cancels later native increments; source-only return does not. The latter can leave a contribution of -1 . The bounded deterministic return check covers 512 words. No infinite stopping or terminal-score deletion theorem is inferred. Decimal cusp examples illustrate a handwritten boundary: support-changing differentiation remains separate from common-support scores.

Sensor availability without an extra PID objective

Assume a finite sensor/target law, an exogenous availability mask M jointly independent of all sensors and target, and an oracle using true conditional probabilities with natural-log loss. The baseline receives the full mask but no sensor readings; the augmented oracle receives that same mask and the available readings. Their optimal losses are $H(T | M)$ and $H(T | M, S_M)$, so their gain is $I(T; S_M | M)$. Joint mask independence reduces each conditional law to the specified sensor/target law. For $q(m) = P(M = m)$, the gain is therefore

$$J = \sum_m q(m) I(S_m; T).$$

The empty subset contributes zero. Self-redundancy and inversion give $I(S_m; T) = \sum_{\alpha \preceq \{m\}} \Pi_\alpha$ for nonempty m . The condition $\alpha \preceq \{m\}$ means that some branch $a \in \alpha$ is contained in m . Interchanging two finite sums therefore gives

$$J = \sum_\alpha \Pi_\alpha w_\alpha, \quad w_\alpha = \sum_m q(m) \mathbf{1}\{\exists a \in \alpha : a \subseteq m\}.$$

Thus this oracle objective is fixed by subset MI; rewriting it in PID coordinates does not supply an extra optimization criterion. Overlapping branches must be counted once. For two independent fair availability bits and singleton branches $\{1\}, \{2\}$, three of four masks qualify: $w = 3/4$, not $1/2 + 1/2 = 1$. Fresh exact checks cover nine laws at two, three and four sources, 75 nonempty subset identities and 18 weighted objectives. They do not prove sampled utility.

Equal synergy can mean different added value

On the same eight equally likely worlds, let A, B, U be independent fair bits, target $T = (A, B)$ and base sensor A . Compare added sensors U, B and A . For U and B , the source union has probability $3/4$ and its target-restricted probability is $1/4$. Redundancy is $\log(4/3)$. Using $\text{Syn} = I(A, C; T) - I(A; T) - I(C; T) + \text{Red}$ gives synergy $\log(4/3)$ in both cases. A duplicate has synergy zero.

Let base and added availability be independent with probabilities $4/5$ and $9/10$, also jointly independent of the bits. Both compared oracles see the same full mask; the baseline gets available base readings, and the augmented oracle also gets available added readings. Added noise has gain zero. The complementary bit is useful whenever it arrives, so its gain is $(9/10) \log 2$. A duplicate is useful only when it arrives and the base fails, so its gain is $(9/10)(1/5) \log 2 = (9/50) \log 2$. Equal synergy therefore does not identify this task's gain.

Pairwise mask independence is insufficient. For independent fair X, Y , set $M = 1$ when $X = Y$. Then M is independent of each individual bit, but not of their joint state. Conditional on observed M , seeing X determines Y . If X is shown only when $M = 1$, its gain given the mask is $(1/2) \log 2$, despite $I(X; Y) = 0$. The exogenous-mask formula cannot be used under this selection.

A finite fault result and real application limits

For the retained binary fault game, set target $T = S_1$, diagonal masses $9/20$ and off-diagonal masses $1/20$. A report must preserve at least one coordinate. Forecast probabilities are strictly positive and sum to one. Worst report loss is averaged under this fixed law.

The admissible source-only attack sends 00 to 01 and 10 with probabilities $4/9, 5/9$, sends 11 with the opposite probabilities, and leaves 01 and 10 unchanged. Each of the four target/report cells has mass $1/4$: $(9/20)(4/9) + 1/20 = 1/4$ and $(9/20)(5/9) = 1/4$. Each surviving report therefore has a fair target.

For positive forecast masses $a + b = 1$, $(a - b)^2 \geq 0$ gives $ab \leq 1/4$. Taking logs gives $(-\log a - \log b)/2 \geq \log 2$. Worst loss dominates the attack's mean, so every forecast has risk at least $\log 2$. Constant fair forecasting attains it. This establishes the finite game's value. The event-consistency forecast is $f_C(t | r) = P(T = t | S_1 = r_1 \text{ or } S_2 = r_2)$. For reports 00, 01, 10 and 11, its target-zero probabilities are respectively $10/11, 10/19, 9/19, 1/11$. Diagonal states have worst correct-target probability $9/19$; off-diagonal states have $1/11$. Their total masses are $9/10$ and $1/10$, so its worst risk is

$$\frac{9}{10} \log(19/9) + \frac{1}{10} \log 11 > \log 2,$$

because both log arguments exceed two. Its risk is about 0.91228249 nats. This refutes the proposed minimax extension, not the published MGW measure. Historical acceptance used ordinary Lean recompilation and axiom output; it was not a separate fresh-kernel campaign.

The physical-intelligence E1 study uses 64 already exposed simulator training episodes and a same-fit quantized empirical law: two camera summaries, pressure and a future target. Earlier compiled Rust, count/log and source-order checks agree. Camera-pair conditional MI given pressure is 0.0072275646778 nats, not the top atom 0.0927952904945. Negative camera atoms remain. The chosen useful-margin comparison failed. No held-out, IID, field or sensor-benefit result follows; this audit did not rerun those compiled studies.

The NEST neural capture has only five lagged rows from six controller steps. One source is a function of another and adds zero conditional MI. Perfect in-table target recovery alone is not forecast performance. A new study needs independent reset trials and a separately fixed external target. In the older office study, a positive empirical conditional-MI increment also coexisted with worse loss for its frozen fitted predictor. The test rows are already exposed; fresh confirmation is required. These negative results stay.

Corrections and next milestones

The foundational descriptor witness cites [Lyu, Clark and Raviv, April preprint version 2](#). The machine registry incorrectly binds that exact witness to their distinct March PRE paper. An isolated two-hunk correction preserves the PRE reference and adds the April source. It remains unapplied: semantic authority must be explicitly reviewed, not merely resealed to pass. A separately cited Matthias source edge is also missing. The guide's cover/navigation says six companions where seven are listed; the existing publication is not fully coherent.

Diagnostic preimages remain: the first dependence test lacked an actual pair law, the first sensor overlap control compared supplied constants, and an unexecuted sensor prototype had an invalid sorting dependency. Their successors use the actual pair, mask law and complete fixed carrier. The first combined readback used the wrong sensor source; it failed closed. Equal-valued wrong integer types exposed another readback weakness and prompted explicit typed checks. None required changing scientific values, proof flags or kernels.

Next scientific milestone: prove the frozen Fin3 carrier/codec/order/registry targets, then separately establish event/count/product/sign semantics, independent solver encoding, directed-log certificates and compiled refinement. Complete proof-body and receipt-stream review remains part of the requested triple audit. Practical sensor and neural work requires predeclared targets, new evaluation data, appropriate dependence handling and non-PID baselines.

No commit, push, main promotion or deletion occurred. Git and outside custody are read-only in this session. All nine named iCloud folders remain protected: whole-root useful-byte successors are unverified, and one historical retrieval failed. Eight roots have no observed allocated file blocks; the remaining root uses about 31 MiB. Existing cloud placeholders are not evidence of duplication.

Evidence and reproduction

Detailed scopes are retained beside this report in `provenance/RESULT.txt`, `formal/RESULT.md`, `formal/ROSTER.md`, the numerical review and its count correction, `numerical/COVERAGE.json` and `statistics/FAILURES.md`. Root readback binds the named finite outputs and their current sources. Reports are preparation, not accepted remote publication or independent storage custody.

Run the prefix and sensor scripts under isolated Python, once normally and once with `-O`. Run the numerical diagnostic and arithmetic scripts with the current integration checkout as their sole argument; the zero-witness script takes none. Then run the scoped `readback.py`. Require actual zero exits, complete records and empty stderr. Same-code output equality is a mode check, not a second scientific derivation. Preserve every rejected output. The PDF build inputs, commands, captured streams and repeatability comparison are retained in separate fresh build directories.